

AI-Powered Smart Student Grievance Management & Resolution System

1. Problem Statement

Traditional college grievance systems often suffer from:

- Manual complaint classification
- Complaints being sent to the wrong department
- Duplicate complaints
- Delayed responses
- Poor visibility of complaint status
- No systematic escalation for overdue complaints
- Difficulty handling complaints written in natural language
- Lack of evidence/document management
- Limited analytics for authorities
- Inconsistent responses from authorities

Proposed Solution

Build an **AI-powered Student Grievance Management System** that receives complaints in natural language, understands and categorizes them, determines priority, detects duplicates, routes them to the appropriate authority, assists with responses, tracks resolution, and automatically escalates unresolved complaints.

The important principle:

*****AI assists understanding and decision-making; human authorities remain accountable for final decisions.*****

2. Main Actors

There should be **three primary roles**.

Student

Can:

- Register/login
- Submit grievance
- Upload evidence
- Track grievance
- Receive notifications
- Respond to authority
- View resolution

- Provide satisfaction feedback
- Appeal/escalate when appropriate

Grievance Authority / Faculty

Can:

- View assigned grievances
- Review AI classification
- Request additional information
- Communicate with student
- Update status
- Add resolution
- Upload supporting documents
- Escalate cases

Administrator

Can:

- Manage departments
- Manage authorities
- Configure grievance categories
- Configure escalation rules
- Monitor system performance
- View analytics
- Audit activities
- Manage users and permissions

3. Complete System Workflow

This should be your **main flow diagram** in the project.

Student

↓

1. Authentication

Secure login → Role identification

↓

2. Grievance Submission

Student enters:

- Title
- Description
- Department
- Location

- Date
- Optional evidence
- Photos/videos/documents

↓

3. AI Grievance Understanding

AI analyzes the complaint and extracts:

- Category
- Sub-category
- Important entities
- Relevant department
- Severity indicators
- Key facts
- Potential urgency

↓

4. Classification

Example:

"The hostel water supply has been unavailable for three days."

AI:

Category: Hostel

Sub-category: Water/Sanitation

Severity: High

Recommended Department: Hostel Administration

↓

5. Duplicate Detection

System compares the complaint against existing complaints using **semantic similarity**.

If similar complaint exists:

"Possible duplicate grievance #GRV-1024 — 91% similarity"

The authority can decide whether to merge/link them.

↓

6. Priority & Severity Engine

Assign:

Low → Medium → High → Critical

based on:

- Complaint category
- Keywords/signals
- Number of affected students
- Safety implications
- Duration
- Historical patterns
- Institutional rules

Important: **don't let the LLM independently decide criticality.** Use a transparent scoring/rule layer with AI signals.

↓

7. Intelligent Routing

System determines:

Complaint → Department → Authority

Example:

Hostel water problem

↓

Hostel Department

↓

Hostel Grievance Officer

↓

8. AI-Assisted Response

AI can generate a suggested response for the authority.

Example:

"Your complaint has been received and forwarded to the Hostel Administration for investigation."

But the authority should **approve/edit before sending** for important cases.

↓

9. Investigation & Resolution

Authority:

- Reviews complaint
- Checks evidence
- Communicates with student
- Takes action
- Uploads resolution evidence
- Changes status

Possible statuses:

Submitted → Under Review → Assigned → In Progress → Resolved → Closed

Also:

Rejected / Need More Information / Escalated / Reopened

↓

10. Escalation Engine

If the complaint remains unresolved beyond its SLA:

Authority → Higher Authority → Administrator

Example:

High priority → 48-hour SLA

No resolution → Automatic escalation

↓

11. Student Feedback

After resolution:

- Satisfaction score
- Feedback
- "Issue resolved?" Yes/No

If unresolved:

Reopen / Appeal

↓

12. Analytics

System generates:

- Complaint volume
- Category distribution
- Department performance
- Average resolution time
- SLA violations
- Duplicate rate
- Satisfaction
- AI accuracy
- Escalation rate

4. AI Layer

This is where your project becomes more than a normal CRUD grievance application.

AI Module 1 — Natural Language Understanding

Input:

"Hostel mein 3 din se paani nahi aa raha aur warden ko complain karne ke baad bhi kuch nahi hua."

System understands:

Category: Hostel

Issue: Water Supply

Duration: 3 days

Previous Action: Warden contacted

Severity: High

Language: Hinglish

This makes the system capable of handling **real student language**, not only predefined forms.

5. AI Module 2 — Automatic Categorization

Suggested categories:

1. Academic
2. Hostel

3. Examination
4. Fees
5. Infrastructure
6. Harassment
7. Transport
8. Library
9. Food & Hygiene
10. Ragging
11. IT/Technology
12. Administration
13. Other

You can create subcategories later.

6. AI Module 3 — Severity Detection

A better model:

LevelExample |

Low | Minor classroom inconvenience

Medium | Repeated administrative issue

High | Major infrastructure/service disruption

Critical | Safety, harassment, ragging, serious misconduct

For critical categories, the system should **prioritize human review rather than blindly automating decisions.**

7. AI Module 4 — Semantic Duplicate Detection

This part of your idea is particularly useful.

Instead of checking exact words:

"Hostel water isn't available."

and

"There has been no water supply in my hostel for two days."

A semantic similarity system recognizes that they may represent the **same underlying issue.**

Better technical approach

Use:

Text → Embedding → Vector Database → Similarity Search

Then:

Similarity > threshold

↓

Possible duplicate

↓

Human confirmation

Don't use an LLM alone for duplicate detection. Embeddings are cheaper, faster, and more measurable.

8. AI Module 5 — Intelligent Routing

Instead of:

Student chooses department manually.

Use:

Complaint Understanding → Category → Department → Authority

Example:

"Exam marks are incorrect"

↓

Category: Examination

↓

Department: Examination Cell

↓

Assigned Authority

This reduces misrouting.

9. AI Module 6 — AI-Assisted Response Generation

The AI shouldn't automatically send arbitrary responses.

Better architecture:

Complaint + Institution Policy + Case Status

↓

LLM

↓

Draft Response

↓

Authority Review

↓

Student

This is much safer and gives you a strong **Responsible AI** angle.

10. Evidence Management

Students should be able to attach:

- Images
- Videos
- PDFs
- Screenshots
- Documents

Each evidence item should have:

- Upload timestamp
- Uploader
- Associated grievance
- Access permissions
- File type
- Audit record

For sensitive evidence, access should be restricted.

11. Notification System

Notifications should occur when:

- Complaint submitted
- Complaint assigned
- Status changed
- Authority requests information
- Response received
- Complaint escalated
- Complaint resolved
- Complaint reopened

Channels:

In-app → Email → Optional SMS/push

12. Escalation Engine

This should be a proper **rule-based subsystem**.

Example:

Complaint Priority = High

SLA = 48 hours

48 hours passed?

↓

YES

↓

Escalate to senior authority

↓

Notify authority + student

For Critical complaints, the escalation window can be much shorter.

13. Security & Responsible AI

Your notes correctly mention this, but it needs to be expanded.

Security

- Authentication
- Role-Based Access Control
- Encryption in transit
- Encryption at rest
- Secure file storage
- Audit logs
- Session management
- Rate limiting
- Input validation

AI Security

You specifically mentioned prompt injection. Good point.

Add:

- Prompt injection detection
- Input sanitization
- System prompt isolation
- Output validation
- Tool/action permission boundaries
- No direct database access for the LLM
- Human approval for sensitive actions

Privacy

Especially important because grievances can contain sensitive information.

Implement:

- Minimum necessary data access
- Role-based evidence access
- Data retention policy
- Audit trail
- Sensitive information protection

14. Responsible AI Principles

This can become a major section of your research paper.

Human-in-the-loop

AI recommends.

Human decides.

Explainability

Instead of:

"Severity = Critical."

show:

"Critical because complaint contains a safety-related issue and indicates immediate risk."

Bias Monitoring

Measure whether the AI behaves differently across:

- Departments
- Complaint categories
- Languages
- Student groups

AI Limitations

The system should be able to say:

"Low confidence — manual review required."

This is much better than forcing an AI prediction every time.

15. Evaluation Matrix

Your handwritten evaluation section is good, but it needs more precise metrics.

A. Classification

Measure:

- Accuracy
- Precision
- Recall
- F1-score

For multiple grievance categories, **macro-F1** is especially useful because some categories may have fewer examples.

B. Routing

Measure:

- Routing accuracy
- Top-1 accuracy
- Top-3 accuracy

C. Duplicate Detection

Measure:

- Precision
- Recall
- F1-score

D. Response Quality

Evaluate:

- Relevance
- Factuality
- Policy compliance
- Human rating

E. Resolution

Measure:

- Average response time
- Average resolution time
- SLA violation rate

F. User Experience

Measure:

- User satisfaction
- Task completion rate
- System usability

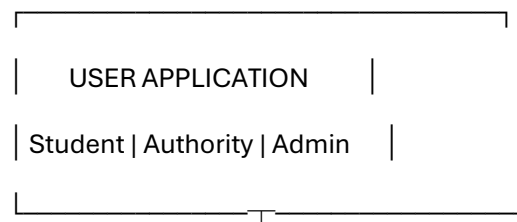
G. Responsible AI

Measure:

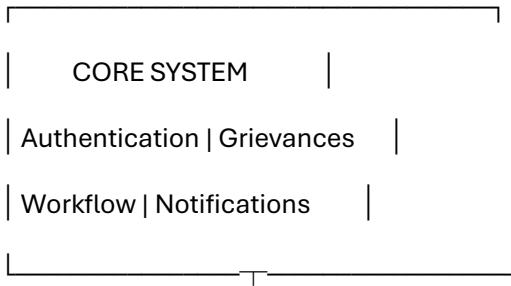
- Hallucination rate
- Human override rate
- Bias/fairness indicators
- Prompt-injection resistance

16. Research-Level Architecture

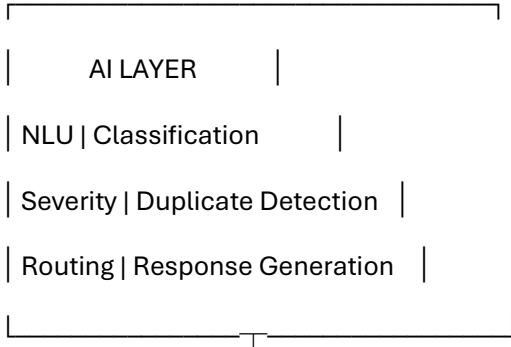
Your system can be divided into **5 layers**:



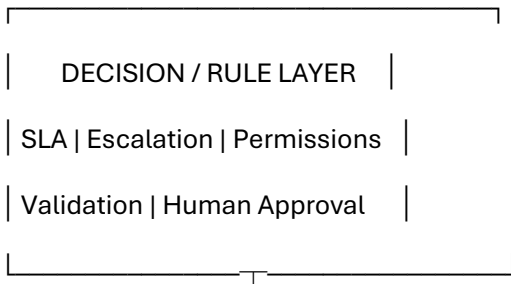
↓



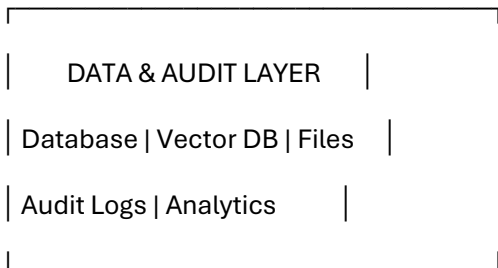
↓



↓



↓



This separation is important because **AI should not directly control critical business operations.**

17. Suggested Technical Stack

For a student project, don't over-engineer it.

Frontend

React + Tailwind CSS

Backend

Node.js + Express

Database

PostgreSQL

Vector Search

pgvector

This is preferable to introducing a completely separate vector database for a relatively small system.

AI

Depending on your experimentation:

- LLM API/local LLM
- Embedding model
- Classification model

Storage

Cloud object storage for evidence.

Authentication

JWT/session-based authentication with RBAC.

Notifications

Email + in-app notifications.

18. What Makes Your Project Innovative?

A basic grievance portal is **not innovative**.

Your innovation comes from combining:

1. Natural-language grievance understanding

Students don't need to understand administrative categories.

2. Semantic duplicate detection

Multiple students reporting the same problem can be linked.

3. Intelligent routing

Complaint automatically reaches the appropriate authority.

4. Dynamic severity assessment

System prioritizes complaints based on content and institutional rules.

5. SLA-based escalation

The system doesn't simply store complaints—it actively tracks unresolved cases.

6. Human-in-the-loop AI

AI assists authorities without replacing accountability.

7. Responsible AI layer

Security, explainability, privacy, bias monitoring and prompt-injection protection.

8. End-to-end analytics

The institution can identify recurring systemic problems.

19. One Important Feature I Would Add

****Complaint Clustering / Issue Detection****

This is stronger than simple duplicate detection.

Suppose:

Student 1 → Hostel water problem

Student 2 → No water in hostel

Student 3 → Hostel taps have no water

Student 4 → Water supply stopped

Student 5 → No water for 3 days

Instead of treating them as five independent complaints:

HOSTEL WATER ISSUE

|

┌──────────┴──────────┐

↓ ↓ ↓

Student 1 Student 2 Student 3...

The system identifies a **campus-wide issue**.

Then the authority sees:

****Emerging Issue: Hostel Water Supply****

37 related grievances

4 hostels affected

First reported: 3 days ago

That turns your project from a complaint-management system into an **institutional problem intelligence system**.

20. Final End-to-End Concept

The cleanest way to describe the project is:

*****An AI-powered student grievance intelligence and resolution platform that transforms unstructured student complaints into categorized, prioritized, intelligently routed, trackable, and measurable cases while keeping human authorities responsible for final decisions.*****

Core pipeline

Student Complaint

↓

Natural Language Understanding

↓

Category + Information Extraction

↓

Severity / Priority

↓

Duplicate + Issue Clustering

↓

Intelligent Routing

↓

Authority Review

↓

AI-Assisted Response

↓

Investigation

↓

Resolution

↓

Student Feedback

↓

Escalation if SLA Violated

↓

Analytics & Institutional Insights

Recommended project structure

For your **25–30 page paper/report**, I would structure it as:

14. ****Introduction****
15. ****Problem Statement & Motivation****
16. ****Literature Review****
17. ****Existing System & Limitations****
18. ****Proposed System****
19. ****System Architecture****
20. ****AI Methodology****
 - NLU
 - Classification
 - Severity
 - Duplicate detection
 - Clustering
 - Routing
 - Response generation
21. ****Workflow & Algorithms****
22. ****Security, Privacy & Responsible AI****
23. ****Implementation / Prototype****
24. ****Evaluation Methodology****
25. ****Results & Discussion****
26. ****Limitations****

27. **Future Work**

28. **Conclusion**

29. **References**

The strongest research angle

Don't present it as merely:

*****"AI Grievance Management System"*****
